

SIMONE D'ONOFRIO



Rome, Italy, 26/7/1997

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EDUCATION

Ph.D., Physics

ICE-CSIC / Universitat Autònoma de Barcelona

2022 – 2026

Barcelona, Spain

Thesis: **Generalized Entropy and $f(R)$ Gravity:**

Cosmological Constraints from Inflation to Dark Energy Submitted, defense expected October 2026

Supervisor: **Sergei D. Odintsov**

M.S., Theoretical Physics - 110L/110

Università La Sapienza

2019 – 2022

Rome, Italy

Thesis: **Classical metrics from quantum amplitudes. The Reissner-Nordström-Tangherlini metric from the graviton emission of charged scalars.** ([Link](#))

Supervisor: **Fabio Riccioni**

B.A., Physics

Università La Sapienza

2016 – 2019

Rome, Italy

Thesis: **Gravitational Love numbers**

Supervisor: **Andrea Maselli**

SKILLS

- **Expertise:** Quantum Field Theories, General Relativity, Theoretical and Observational Cosmology, Modified Gravity Theories, Compact Objects, Numerical Simulations, Gravitational Waves, Gravity as an Effective Field Theory, Tidal deformability
- **Computer:** Deep knowledge of C, C++, Python, Mathematica (including xAct package), Shell Scripting, LaTeX, RStudio, C#. Ability to use computing clusters and parallelize programs.
- **Languages:** English (fluent), Spanish (fluent), Italian (native) and Catalan (basic).

PUBLICATIONS

- [1] Simone D'Onofrio, Sayak Datta, and Andrea Maselli. “Axial tidal Love numbers of black holes in matter environments”. In: (May 2026). arXiv: 2605.02633 [gr-qc].
- [2] Simone D'Onofrio, Dong Ha Lee, Eleonora Di Valentino, and Juan García-Bellido. “General Relativistic Entropic Acceleration at the perturbation level: a CLASS implementation and first Boltzmann-code constraints”. In: (July 2026). arXiv: 2607.25841 [gr-qc].
- [3] Simone D'Onofrio, Sergei Odintsov, and Tanmoy Paul. “Fitting NANOGrav 15-year data and ACT data with modified inflation in entropic cosmology”. In: *Phys. Rev. D* 113.4 (2026), p. 043527. DOI: 10.1103/ktt2-lm19. arXiv: 2510.20484 [gr-qc].
- [4] Simone D'Onofrio, Sergei Odintsov, and Tiziano Schiavone. “Exponential $f(R)$ cosmology with massive neutrinos as a dynamical dark energy framework”. In: *Annals Phys.* 494 (2026), p. 170621. DOI: 10.1016/j.aop.2026.170621. arXiv: 2511.06924 [gr-qc].
- [5] Simone D'Onofrio. “I-Love-Q relations for neutron stars with dark energy”. In: *Eur. Phys. J. C* 85.6 (2025), p. 672. DOI: 10.1140/epjc/s10052-025-14397-2. arXiv: 2506.01889 [gr-qc].
- [6] Sergei D. Odintsov, Simone D'Onofrio, and Tanmoy Paul. “Generalized entropic dark energy with spatial curvature”. In: *Phys. Dark Univ.* 48 (2025), p. 101920. DOI: 10.1016/j.dark.2025.101920. arXiv: 2504.03470 [gr-qc].
- [7] Sergei D. Odintsov, Simone D'Onofrio, and Tanmoy Paul. “Entropic Inflation in Presence of Scalar Field”. In: *Universe* 10.1 (2024), p. 4. DOI: 10.3390/universe10010004. arXiv: 2312.13587 [gr-qc].

- [8] Sergei D. Odintsov, Simone D’Onofrio, and Tanmoy Paul. “Primordial gravitational waves in horizon cosmology and constraints on entropic parameters”. In: *Phys. Rev. D* 110.4 (2024), p. 043539. DOI: 10.1103/PhysRevD.110.043539. arXiv: 2407.05855 [gr-qc].
- [9] Simone D’Onofrio. “Holographic description of F(R) gravity coupled with axion dark matter”. In: *Int. J. Geom. Meth. Mod. Phys.* 20.13 (2023), p. 2350224. DOI: 10.1142/S0219887823502249. arXiv: 2306.16743 [gr-qc].
- [10] Sergei D. Odintsov, Simone D’Onofrio, and Tanmoy Paul. “Holographic realization from inflation to reheating in generalized entropic cosmology”. In: *Phys. Dark Univ.* 42 (2023), p. 101277. DOI: 10.1016/j.dark.2023.101277. arXiv: 2306.15225 [gr-qc].
- [11] Simone D’Onofrio, Federica Fragomeno, Claudio Gambino, and Fabio Riccioni. “The Reissner-Nordström-Tangherlini solution from scattering amplitudes of charged scalars”. In: *JHEP* 09 (2022), p. 013. DOI: 10.1007/JHEP09(2022)013. arXiv: 2207.05841 [hep-th].
- [12] Simone D’Onofrio. “Classical metrics from quantum amplitudes: The Reissner-Nordström-Tangherlini metric from the graviton emission of charged scalars”. MA thesis. Rome U., 2021.

CONFERENCES

Dark Matter and Stars (ICDMS) **13–15 Jul 2026**

University of Southampton. 20-minute contributed talk. Fee waiver awarded. Talk title: Tidal Love Numbers of Black Holes Dressed by Dark Matter.

PASCOS 2026 **22–26 Jun 2026**

String/Black Hole parallel session. University of Sheffield. 20-minute presentation. Fee waiver awarded. Talk title: Tidal Love Numbers of Black Holes Dressed by Dark Matter.

2nd Int. Conference on the Physics of the Two Infinities **17–21 Nov 2025**

Cosmology parallel session. University of Tokyo. 15-minute presentation. Funded by ILANCE-CNRS.

6th BIG&C Meeting **4 Jul 2025**

ICE-CSIC, Barcelona. 20-minute presentation.

Gravity 2025 Conference **27–31 Jan 2025**

YITP. Yukawa Institute for Theoretical Physics, Kyoto. Poster presentation.

GSSI–JHU Meeting **16–18 Dec 2024**

Assergi, Italy. 15-minute talk.

SCIENTIFIC ACTIVITIES

Invited Seminar at La Sapienza University, Rome **23 Sep 2026**

One-hour invited seminar (host: Paolo Pani). Talk title: Tidal deformability of compact objects and the effects of an environment.

Invited Seminar at Queen Mary University of London **8 Jul 2026**

One-hour invited seminar. Talk title: Tidal deformability of compact objects and the effects of an environment.

Invited Seminar at Keio University, Tokyo **17 Nov 2025**

20-minute invited seminar (host: Antonino Flachi).

Invited Seminar at Università degli Studi di Cagliari **12 Nov 2025**

20-minute invited seminar (host: Riccardo Murgia).

Refereeing activities

I have refereed more than ten papers for various journals, among which Physics of the Dark Universe and JHEAP.

VISITING EXPERIENCES

- Invited visiting student to University of Sheffield** 28/09 - 9/10 2026
With Prof. Eleonora Di Valentino and funded by the STSM project of the COST Action CA21136
- Visiting student to University of Sheffield** June–July 2026
With Prof. Eleonora Di Valentino
- Invited visit to Università degli Studi di Cagliari** 8-14/11/2025
With Prof. Riccardo Murgia
- Invited visit to GSSI institute** 9-13/6/2025
With Prof. Andrea Maselli
- Visiting student to GSSI institute** November–December 2024
With Prof. Andrea Maselli
- Visiting student at Università Federico II of Naples** September–October 2023
With Prof. Mariafelicia De Laurentis

OUTREACH ACTIVITIES

(All of the following activities were performed during my PhD in Spain and all of them have been done in Spanish.)

- MAGNET Project** 2023 – 2025
I was part of the Magnet project which consists of a Spanish campaign to link research institutes to primary schools with social problems. I organized and carried out various activities with classes of young kids about various topics of physics during all the duration of the project.
- Workshop at *Researchers' night* in UAB** 26/9/2025
Organised an activity at the UAB University for the Resercher's night in which we presented basic concepts of General Relativity to the participants of the event.
- Workshop at *Festa de la Ciencia* in Barcelona** 31/5/2025
Together with Jorge Riviero, we organised an activity at the science fair of the municipality of Barcelona named "Life and death of stars", in particular I explained the formation of black holes and their properties.
- Lecture *Ciencia en el Barrio*** 20/3/2025
Together with Michele Lenzi, we organised a lecture with the CSIC (Spanish national research institute) to explain General Relativity and Cosmology to students of the High School "IES Miquel Tarradell" of Barcelona. The students then went to the National Fair of "Ciencia en el Barrio" in Sevilla (one of the biggest outreach events in Spain) to present what they learned from the activity.
- Lecture *Ciencia en el Barrio*** 14/3/2025
Together with Michele Lenzi, we organised a lecture with the CSIC (Spanish national research institute) to explain General Relativity, Black Holes and Gravitational Waves to students of the High School "Institut Front Marítim" of Barcelona.
- Workshop at *Resercher's night* in UAB** 27/9/2024
Together with Michele Lenzi, we organised an activity at the UAB University for the Resercher's night in which we presented basic concepts of General Relativity to the participants of the event.
- Workshop *Ciencia en el Barrio*** 9/6/2024
Together with Jorge Riviero, we organised a workshop with the CSIC (Spanish national research institute) to explain the Dark Universe to the people of the city.
- Workshop at *Saló de l'Ensenyament*, Barcelona's annual fair** 13/3/2024
Together with Jorge Riviero, we organised a workshop on Black Holes at the Barcelona's annual university fair.